

Introduction to AI Review

Artificial Intelligence

1. What is intelligence?

Solution: Intelligence is the ability to do:

- **Problem solving:** transforming a given state into a goal state.
- **Inference:** deriving new knowledge from existing knowledge.
- **Reasoning:** answering questions or justifying answers.
- **Decision making:** choosing actions that maximize expected utility.
- **Learning:** using experience (e.g., data) to get better at a task, such as those listed above.

2. What is artificial intelligence?

Solution: A human-made (artificial) system that does any of the above is an AI system.

3. What are the four approaches to AI?

Solution:

4. Explain “thinking humanly.”

Solution:

5. Explain “acting humanly.”

Solution:

6. What is the Turing Test?

Solution: A computer is intelligent if a human interrogator, after posing written questions, can't tell whether written responses come from a person or from the computer.

7. Explain “thinking rationally.”

Solution:

8. Explain “acting rationally.”

Solution:

9. When was the term “Artificial Intelligence” coined, and by whom?

Solution: John McCarthy coined the term artificial intelligence in 1955 in his proposal for the first workshop on AI to take place at Dartmouth College in the summer of 1956.

10. Geoff Hinton has been called the “godfather of AI.” How old was Geoff Hinton when the field of AI was launched?

Solution: 8

11. What is the **strong AI** hypothesis?

Solution: The assertion that machines that execute intelligent behavior are actually consciously thinking.

12. What is the **weak AI** hypothesis?

Solution: The idea that machines can *act* as if they are intelligent, that is, they can *simulate* intelligent behavior.

13. What is **narrow AI**?

Solution:

14. What is **general AI**?

Solution:

15. Is artificial intelligence possible?

Solution: